



MBM Engineering College Jodhpur
Department of Mechanical Engineering

***Department
Of
Mechanical Engineering***



LAB MANUAL

Thermal Engineering

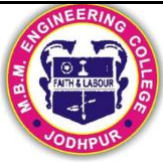
Laboratory-IV (272B)

III year/ VI Semester

2020-2021

Program Outcomes:

- PO1 **Engineering knowledge:** Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
- PO2 **Problem analysis:** Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
- PO3 **Design/development of solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
- PO4 **Conduct investigations of complex problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
- PO5 **Modern tool usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
- PO6 **The engineer and society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
- PO7 **Environment and sustainability:** Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
- PO8 **Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
- PO9 **Individual and team work:** Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
- PO10 **Communication:** Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
- PO11 **Project management and finance:** Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
- PO12 **Life-long learning:** Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.



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Vision of the Institution

“To be a leading educational institute that provides quality technical education and conducts research to produce knowledge-rich professionals for meeting the dynamic needs of the industry and society.”

Mission of the Institution

1. To impart quality technical education to the students to make them globally competent engineers, contributing to the development of the nation and world at large.
2. To imbibe ethical values, scientific and industrial temperament, and spirit of innovation among students.

Vision of the Department

To provide quality technocrat compatible with global standard in the field of mechanical engineering who can contribute to society through innovations, entrepreneurship & sustainable development.

Mission of the Department

- M1. To impart highest quality in fundamentals, technical knowledge & scientific education to the learners to make them globally competitive mechanical engineers.
- M2. To promote liaison with world class industries & educational institutions for excellence in teaching, research & consultancy practices.
- M3. To make students, life-long learners, ethically and technically capable to build their careers in terms of professions and personality.

Program Specific Outcomes:

- PSO1 To empower the students to understand the basic concepts of mechanical engineering. In addition to this, application of theoretical and practical skills to solve complex mechanical engineering problems with the use of advanced software tools to arrive at appropriate and cost effective solutions.
- PSO2 To develop entrepreneurship skills compatible with industrial readiness and can pursue higher studies in Mechanical Engineering or interdisciplinary programs with the concern for sustainable development of society & high regard of ethical values.



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Course Outcomes (Cos)

ME 272 B Thermal Engineering Laboratory –VI (ME)	CO1	Practical exposure to advances in automobile engineering and I.C. engines
	CO2	To develop the interest of students in automobile engineering.

VI Semester: ME

Course Code	Category	Hours / Week			Credits	Maximum Marks		
		L	T	P		Pract	Sess	Total
ME 272-B	Core	-	-	3	1.5	100		100
Lectures Classes: Nil	Tutorial Classes: Nil	Practical Classes: 36				Total Classes: 36		

COURSE OBJECTIVES:

1. To provide an insight into Automobile functioning.
2. To provide an insight into different systems and equipment being used in I.C. Engines.
3. To provide training to students to enhance their practical skills.
4. To develop team qualities and ethical principles.

Safety:

1. To avoid injury, the student must take the permission of the laboratory staff before handling any machine. Careless handling of machines may result in serious injury.
2. Students must ensure that their work areas are clean and dry to avoid slipping.
3. A leather apron will be issued to each student during Welding Exercise. Students not wearing the apron will not be permitted to work in the laboratory. Students are required to clear off all tools and materials from machine/work place.
4. At the end of each experiment, students must clear off all tools and materials from the work area.
5. Use the appropriate machine for the work to be done.
6. Safety shoes must be worn inside workshop.



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List of Experiments

Sr. No.	Date	Name of Experiments	Page No.	Staff Signature
1.		To Study various automobile emissions and their norms and control system.		
2.		To study Engine starting motor (D.C. Motor) and driving arrangement (Benedix Drive).		
3.		To study DC generator and Alternator storage battery charging drive.		
4.		To Study advanced fuel injection system: ECM - Electronic Control Module MPFI - Multi Point and injection system CRDI- common rail direct injection system. DICOR- Direct injection common rail system.		
5.		To study various types of carburetor: S.V. Carburetor Zenith Carburetor Solex Carburetor		
6.		To study various cooling methods. (i) Air cooling (ii) Water cooling (iii) Liquid coolant(radiator) cooling		
7.		To study various lubrication systems (splash, pressure feed, mechanical)		
8.		To study advanced ignition system for SI engines.		
9.		To study advanced valve timing system of a modern passenger car.		
10.		To perform Morse test on Petrol engine and draw performance curve.		



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EXPERIMENT NO. 1: TO STUDY VARIOUS AUTOMOBILE EMISSIONS: THEIR NORMS AND CONTROL SYSTEM.

Introduction

Vehicles on road are the key contributors to a nation facilitating timely movement of goods and other consumer-related activities. However, rising air pollution levels have posed a challenge in developing countries with severe problems of congestion and growing vehicular movement.¹ This has led to cause many problems like:

- Global warming
- Respiratory disorders (Asthma)
- smog
- Acid rain
- Premature skin aging and discoloration

Theory:

Internal Combustion Engines (IC Engines) have been around only for a fraction of human history but have led to drastic changes not just in the terms of complexity of engineering but also because of by-products obtained due to their use. A CO₂ molecule has a lifetime of about 1000 years, so that means with first kick to start your bike in the morning, you gave this planet some molecules for the next millennia. In this Experiment we will look at different aspects of what technology has made a difference in the domain of IC engines concerning Environment and what norms and policies of emission have been decided by the National and International committees to deal with Air pollution and environmental degradation in general.

Spark Ignition Engine Emission Control

SI engine vehicles without emission control have three sources of emissions.

- Exhaust emissions
- Crankcase emissions
- Evaporative Emissions



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NO_x and CO are formed in the burned gases in the cylinder. Unburned HC emissions originate when fuel escapes combustion due to several processes such as flame quenching in narrow passages present in the combustion chamber and incomplete oxidation of fuel that is trapped or absorbed in oil film or deposits

- NO_x is formed by oxidation of molecular nitrogen. During combustion at high flame temperatures, nitrogen and oxygen molecules in the inducted air breakdown into atomic species which react to form NO. Some NO₂ is also formed and NO and NO₂ together are called as NO_x.

- CO results from incomplete oxidation of fuel carbon when insufficient oxygen is available to completely oxidize the fuel. CO rises steeply as the air-fuel (A/F) ratio is decreased below the stoichiometric A/F ratio.

- HC originates from the fuel escaping combustion primarily due to flame quenching in crevices and on cold chamber walls, fuel vapour absorption in the oil layer on the cylinder and in combustion chamber deposits, and presence of liquid fuel in the cylinder during cold start.

Air-fuel ratio is one of the most important parameters that affect the engine exhaust emissions. The SI engine is operated close to stoichiometric air-fuel ratio as it provides a smooth engine operation. Nitric oxide emissions are maximum at slightly (5-10 %) leaner than stoichiometric mixture due to combination of availability of excess oxygen and high combustion temperatures at this point. Carbon monoxide and HC emissions reduce with increase in the air-fuel ratio as more oxygen gets available for combustion. Lean engine operation to a certain critical value of air-fuel ratio tend to reduce all the three pollutants. Further leaning of mixture results in poor quality of combustion and eventually in engine misfiring causing an erratic engine operation and sharp increase in HC emissions. Normally, one would like to operate engine on lean mixtures that would give low CO and HC, and moderate NO_x emissions. But, presently most engines are operated very close to stoichiometric conditions for catalytic control of NO_x emissions.

Compression ignition engine emission control

A fully developed diesel spray may be considered to consist of three distinct regions based on the variations in fuel-air equivalence ratio ϕ , across the cross section of the spray as seen radially outwards from the centerline of spray.



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- A fuel rich core where fuel-air equivalence ratio is richer than the rich flammability limits i.e. $\phi > \phi_R$.
- Flammable region in which ϕ lies within the rich and lean flammability limits, i.e. $\phi_R > \phi > \phi_L$
- A lean flame-out region (LFOR) where ϕ is lower than lean flammability limits and extends up to the spray boundary i.e. $\phi_L > \phi > 0$.

Pollutant formation is strongly dependent on the fuel air ratio distribution in the spray

- NO is formed in the high temperature burned gases in the flammable region. Maximum burned gas temperatures result close to stoichiometric air-fuel ratio and these contribute maximum to NO formation.
- CO is formed in fuel rich mixtures in the flammable region.
- Soot forms in fuel-rich spray core where fuel vapor is heated by the hot burned gases.
- Unburned HC and oxygenated hydrocarbons like aldehydes originate in the region where due to excessive dilution with air the mixture is too lean at the spray boundaries. In excessive lean mixtures combustion process either fails to begin or does not reach completion. Towards the end of combustion, fuel in the nozzle sac and orifices gets vaporized, enters the combustion chamber and contributes to HC emissions.

An overview and control of engine-out emissions

The first emission control for the spark ignition engines involved adjustments of air-fuel ratio. It was followed by control and adjustment of other engine parameters such as mixture control under idling, acceleration and deceleration, spark timing, precision manufacturing of key engine components such as piston, rings, cylinder head gasket to minimize crevice volume, cams, valves etc.

Positive crankcase ventilation (PCV) system was introduced on gasoline vehicles during mid 1960's to prevent release into atmosphere of hydrocarbon-rich crankcase blow by gases as the emission standards were tightened, exhaust aftertreatment devices such as catalytic converters



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were introduced for the first time in 1974-75 and more advanced modifications in engine design and fuel system were employed. Electronic fuel and engine management become necessary during 1980s to meet the then emission regulations. Further advancements in engine, fuel system and emission control technology have emerged in the meantime.

Multi-valve cylinder engines became common and variable valve actuation was applied in production vehicles during late 1980s. In mid 1990s, gasoline direct injection stratified charge (DISC) engines were put into production by Japanese auto-manufacturers. So, today the emission control techniques may be grouped into the following broad categories:

Engine design and fuel system parameters

- Engine compression ratio,
- Combustion chamber design – low crevice volume, high turbulence
- Spark timing
- Air-fuel ratio
- Fuel system design: carburetor giving way to fuel injection
- Multivalve and variable valve actuation
- Engine temperature control

Engine add-on(s) to enable reduction of engine-out emissions

- Crankcase Emission Control (PCV System)
- Evaporative Emission Control
- Exhaust Gas Recirculation
- Water Injection

Exhaust after-treatment

- Thermal reactors
- Catalytic Reactors



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Exhaust gas re-circulation

The heat capacity of the exhaust gas is higher than the air as it contains significant amount of tri-atomic gases CO₂ and water vapors. Therefore, addition of exhaust gas to fresh intake charge has a higher effect in lowering the combustion temperatures compared to simple leaning of the charge. EGR is defined as a mass percent of total intake flow:

$$EGR = \frac{\dot{m}_{EGR}}{\dot{m}_i} \times 100$$

where, $\dot{m}_i$ = Total mass flow into the Engine. Typically, only about 5 to 10 % EGR rates are employed. At higher EGR rates, frequency of partial and complete misfire cycles increases resulting in unacceptably higher HC emissions and loss in fuel economy and power. EGR systems are made to operate mostly in the part-load range.

These are deactivated at engine idle, because large amount of residual gas is already present in the cylinder. Many times, the system is deactivated at full throttle conditions as the vehicle rarely operates under these conditions during city operation. A schematic layout of EGR system is shown in Fig 5.8. An EGR control valve is used to regulate flow of EGR depending upon engine operating conditions.

The intake manifold pressure or exhaust back pressure may be used to control EGR rate as these parameters vary with engine load. In the modern engines, EGR rate is controlled by the engine electronic control unit. A pressure sensor in the exhaust or intake provides signal to the electronic control module of the engine, which in its turn regulates the operation of the EGR valve. Electronically controlled EGR valves actuated by high-response stepper motor are being used on modern engines.

Their fast response during transient operation makes it possible to reduce NO_x more than what is obtained by use of a mechanically controlled EGR valves. Effectively a lower rate of EGR can be employed to obtain the same reduction in NO_x that results in a lower penalty on HC emissions.

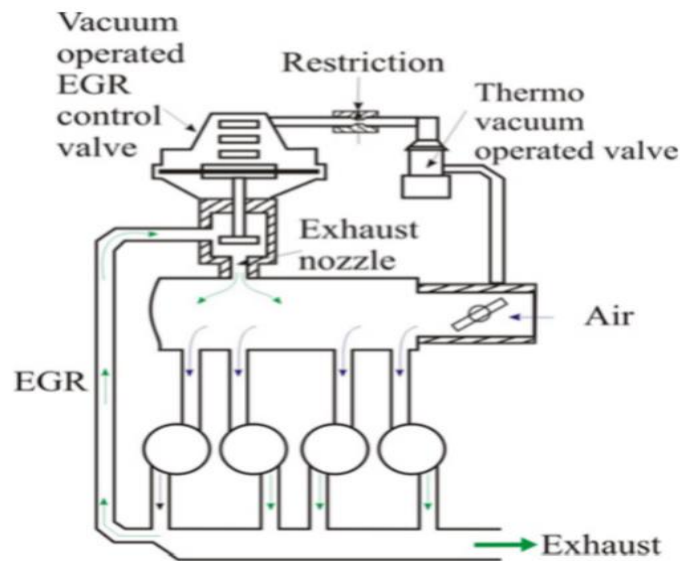


Fig1.1: EGR System

Catalytic converter

They belong to the category of Exhaust After treatment. Improvements in engine design and adjustment of engine parameters carried to control engine emissions were inadequate to meet the first set of stringent emission regulations introduced in the USA from 1975. Devices were developed to treat exhaust gas for conversion of engine emissions to harmless gases. Conversion of pollutants present in the exhaust gas is termed as 'exhaust after treatment'. Two basic types of exhaust treatment systems were considered:

- Thermal reactors
- Catalytic Converters

today, catalytic converters have become a standard exhaust aftertreatment device for the spark ignited engine vehicles. Catalytic converters are now a standard fitment to SI engine powered vehicles. The exhaust gas flows over a bed of catalyst where pollutants are converted to harmless gases. The catalyst lowers the reaction temperature and hence high conversion rates are obtained compared to thermal reactors. A catalytic converter consists of the following main elements besides housing:

1. Catalyst

- *Palletized Catalysts*



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- *Ceramic Monoliths*
- *Metallic Monoliths*

2. Catalyst Substrate or support

3. Intermediate coat or wash coat

Different types of Catalytic converters in the production gasoline vehicles two main types of catalytic converters have been used:

Oxidation catalytic converter: The oxidation catalyst converts CO and HC to CO₂ and H₂O at substantially low temperatures and at higher conversion efficiency than the thermal reactors. The required oxygen for oxidation reactions is made available either by operating engine lean or by injecting secondary air ahead of the catalytic converter when engine is operated rich. Vehicles employing the oxidation catalysts were generally tuned rich for better NO_x control and secondary air injection was employed.

Three-way catalytic converter: The essential condition to simultaneously oxidize CO and HC and reduce NO_x on the same catalyst bed is to operate engine at very close to stoichiometric air-fuel ratio. Under stoichiometric engine operation, enough reducing gases CO and HC are present in the exhaust to reduce NO_x to N₂ and at the same time enough oxygen to oxidize CO and HC. Thus, all the three pollutants are removed simultaneously in a 3-way catalyst.

Reduction and 3-way catalytic converters: For reduction of NO_x, the CO, HC and H₂ which are present in the exhaust during rich engine operation are made to react with NO_x over a catalyst when NO_x is converted to N₂.

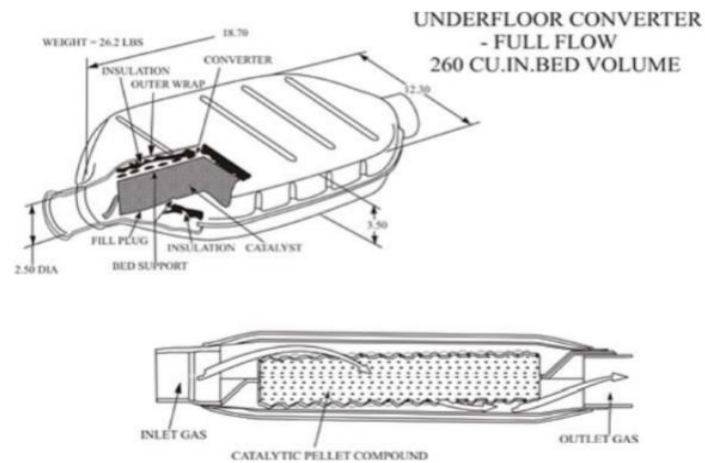


Fig 1.2: Underfloor Converter

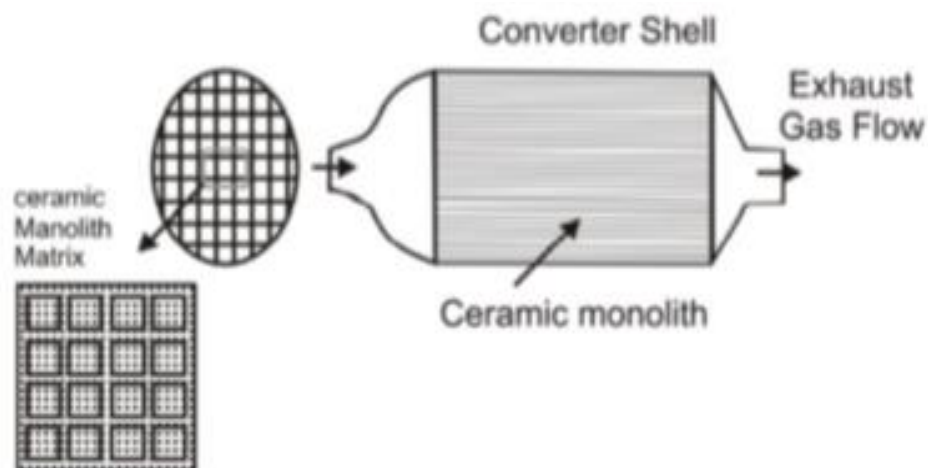
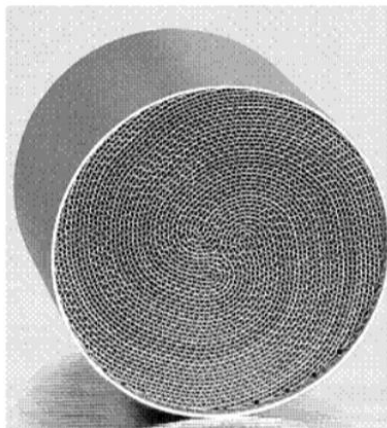


Figure1.4: Ceramic monolith catalyst substrate having square cells.



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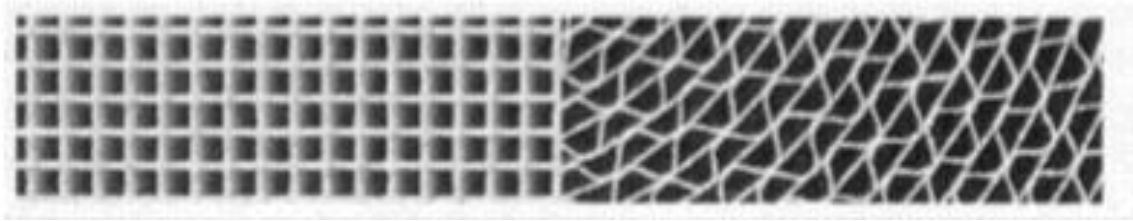


Fig1.5: Comparison of Ceramic monolith with metallic monolith

Bharat stage VI Norm

Bharat Stage norms are the automotive emission norms which the automotive manufacturers have to comply to sell their vehicles in India. These norms are applicable to all two wheelers, three wheelers, four-wheelers and construction equipment vehicles. To curb growing menace of air pollution through the vehicle's emission, the Government of India has decided to leapfrog from the exiting BS – IV norms to the BS- VI, thereby skipping the BS – V norms, and to implement the BS – VI norms with effect from 1st April 2020. Only those vehicles will be sold and registered in India from 1st April 2020 onwards, which comply to these norms. The norms are stringent and as par with global standards.

International Centre for Automotive Technology (ICAT) has released India's first Type Approval Certificate (TAC) for Bharat Stage – VI (BS – VI) norms for the two wheeler segment. Last year, ICAT issued the approval for BS –VI norms to M/s Volvo Eicher Commercial Vehicles for the Heavy Commercial Vehicle segment which was also the first in its segment in India. ICAT is the premier testing and certification agency authorized by Ministry of Road Transport and Highways for providing testing and certification services to the vehicle and component manufacturers in India and abroad. Upgrading to stricter fuel standards helps tackle air pollution. Global automakers are betting big on India as vehicle penetration is still low here, when compared to developed countries. At the same time, cities such as Delhi are already being listed among those with the poorest air quality in the world. The national capital's recent odd-even car experiment and judicial activism against the registration of big diesel cars shows that governments can no longer afford to relax on this front. With other developing countries such as China having already upgraded to the equivalent of Euro V emission norms a while ago, India has been lagging behind. The experience of countries such as China and Malaysia show that poor air quality can be bad for bad for business. Therefore, these reforms can put India ahead in the race for investments too.

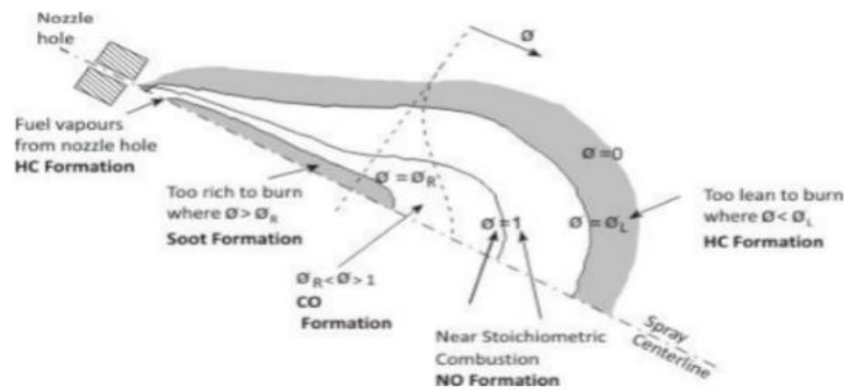


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Difference between BS-IV and the new BS-VI

1. The major difference in standards between the existing BS-IV and the new BS-VI auto fuel norms is the presence of Sulphur.
2. The newly introduced fuel is estimated to reduce the amount of Sulphur released by 80
3. As per the analysts, the emission of NO_x (nitrogen oxides) from diesel cars is also expected to reduce by nearly 70.

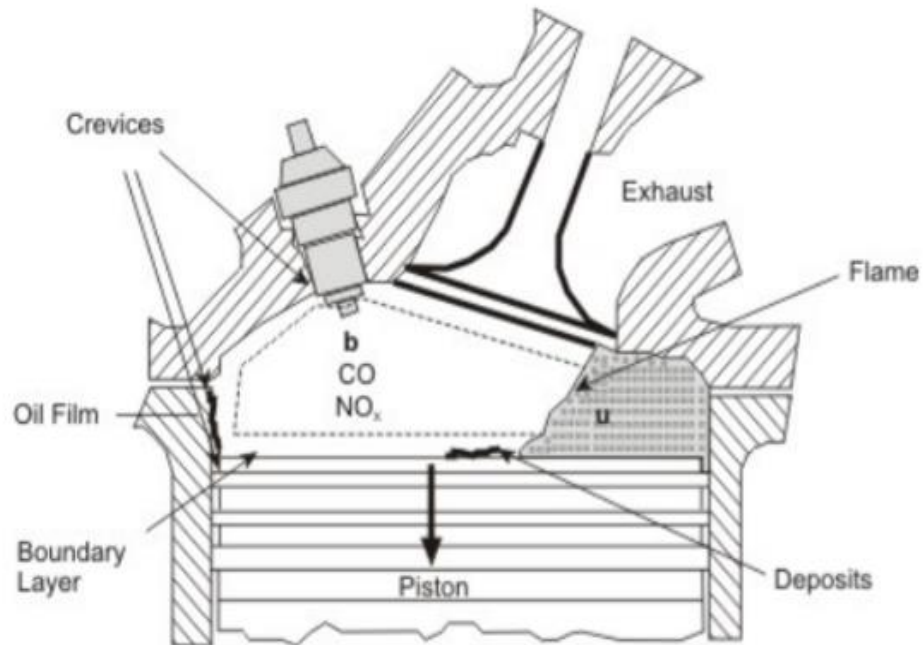
Schematics:



1.	$\text{NO} + \text{CO} \rightarrow \frac{1}{2} \text{N}_2 + \text{CO}_2$
2.	$\text{NO} + 5\text{CO} + 3\text{H}_2\text{O} \rightarrow 2\text{NH}_3 + 5\text{CO}_2$
3.	$2\text{NO} + \text{CO} \rightarrow \text{N}_2\text{O} + \text{CO}_2$
4.	$\text{NO} + \text{H}_2 \rightarrow \frac{1}{2} \text{N}_2 + \text{H}_2\text{O}$
5.	$2\text{NO} + 5\text{H}_2 \rightarrow 2\text{NH}_3 + 2\text{H}_2\text{O}$
6.	$2\text{NO} + \text{H}_2 \rightarrow \text{N}_2\text{O} + \text{H}_2\text{O}$



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EXPERIMENT NO6: TO STUDY VARIOUS AUTOMOBILE COOLING METHODS.

Object: To study various cooling methods.

- (i) Air cooling
- (ii) Water cooling
- (iii) Liquid coolant(radiator) cooling

Theory:

Introduction: Only a part of the total fuel energy supplied to the internal combustion engine is converted into useful work; about 30% is lost as exhaust heat and some energy is left in friction compensation and direct rejection from engine.

The rest of the energy about 30% has to be removed by the cooling system if it is not increasing the engine temperature while running.

Types of cooling system: As the heat rejected from the engine ultimately goes to air. Nevertheless, two basic systems are used to cool the engine.

They are: -

- (i) Air cooling
- (ii) Water cooling or indirect air-cooling using water as a transfer medium.

Air cooling

Air cooling is used for small engine and engine whole applications gives extreme importance to weigh such as air craft engines. For air cooling of the engine the heat transfer area is increased by timing the cylinder and cylinder head and air is passed over them. The length of fins and spacing between them is quite important for efficient cooling. Larger interspacing offers larger area for cooling air but the heating of air is less, so more cooling air is required. Smaller interspacing between the fins results in smaller flow area for cooling air and hence air is heated more as input to cooling is air is less. The upper part of cylinder barrel is not critical part of the air-cooled engine and usually limits the engine rotary.

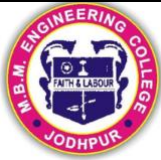
Water cooling

In case of water-cooled engines, the cylinder head and the cylinder are closed in a water jacket. Water is circulated through this jacket to affect cooling. The local velocity of water as well as the bulk rate of water passed through the jacket are important (greater than 3-4 /sec).

For ease of cold starting, many types and antifreeze solutions are added to the water.

These are:

1. Kerosene
2. Wood alcohol
3. Denaturized Alcohol
4. Glycerin
5. Sugar solution
6. Calcium or magnesium chloride
7. Ethyl glycol and Propylene Glycol.



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Various methods are used for circulating the water around the cylinder and cylinder head.

These are:

- (i) Thermo syphon cooling
- (ii) Forced or pump cooling
- (iii) Cooling with thermostat regulation
- (iv) Pressurized water cooling
- (v) Evaporative cooling

Thermostat cooling: A thermostat (a temperature cooling device) is used to stop flow of a coolant below a present cylinder barrel temperature.

A thermostatic device which prevents the water in the engine jackets from circulating through the radiator for cooling until its temperature has reached a value suitable for efficient engine operation.

The thermostat bypass valve is filters between the engine and the radiator top.

Advantage of air cooling:

- (i) Direct passage of heat from engine to air eliminates the use of water as a coolant.
- (ii) Air cooled engine are less sensitive to climate conditions.
- (iii) Control of cooling systems is much easier.
- (iv) Warm up performance is better.
- (v) Engine design is simpler.
- (vi) No water jacket radiator and water pumps are required

Disadvantage of air-cooling system:

- (i) Due to absence of water passage the combustion noise is not attended.
- (ii) Volumetric efficiency is lower due to higher cylinder head temperature.
- (iii) High specific output engine cannot be air cooled due to complex nature to the fins that are required.

Advantage of water-cooled engine:

- (i) Water cooled engine can be installed anywhere in vehicle.
- (ii) Volumetric efficiency is higher.
- (iii) Compact design.
- (iv) High specific output engine pores no problem with water cooling.

Disadvantages of water systems:

1. Need for a radiator and a pump increases the weight and the dimensions of the engine.
2. Water cooling systems requires more maintenance.
3. Power absorbed by the pump is slightly higher than the necessary for air cooled systems.
4. Warm up performance is poor.

Radiators: The function of the radiator is to reject coolant heat the outside air.

The cooling effect in a radiator is achieved by dispensing the heated coolant into fire systems



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through the radiator matrix so that relatively small quantities of coolant are brought in contact with large metal surface then surface of metal to air.

The material used for radiations must have good corrosion resistance in addition to good thermal conductivity.

Example: Yellow brass, Copper.

Radiator headers are also made of copper until the supporting sides are usually made of steel.

Cooling fans: The purpose of providing a fan is to maintain as adequate air flow across the radiator matrix especially at low-cost speeds under engine idling condition.

Any at high speeds can the cam effect provide all the necessary air flow.

It is designed to cool the engine at the lowest possible fan speed when the engine is operating at its highest coolant temperature.

This requires air ducting arrangement and fan design etc.

Schematics:

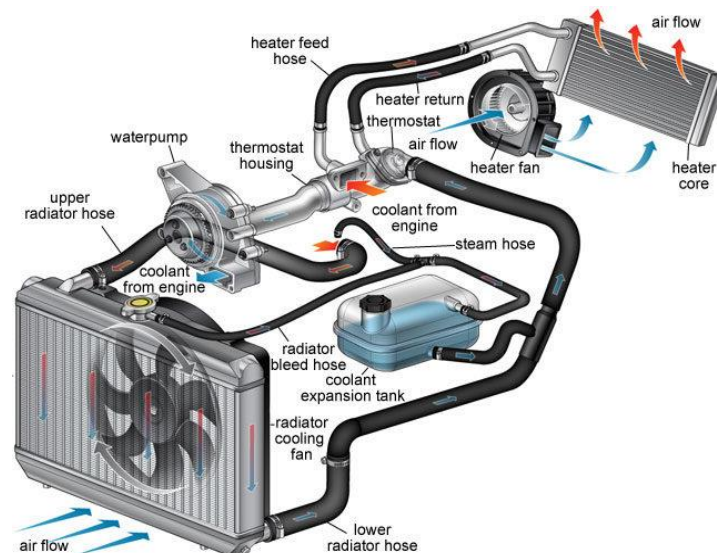


Figure6.1: Automobile Radiator



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EXPERIMENT NO4: To study advanced fuel injection system

Object: To Study advanced fuel injection system:

ECM - Electronic Control Module

MPFI - Multi Point and injection system

CRDI- common rail direct injection system

DICOR- Direct injection common rail system

Theory: -

Drawbacks of modern carburetor

The mixture supplied to various cylinders of multi cylinder engine varies in quality and quantity. also due to fuel condensation in induction manifold, the mixture proportions is affected. – Due to pressure of several wearing parts, the carburetors operate at lower efficiency. – Reduced volumetric efficiency due to non-availability of free flow parole for the reactive mixture owing to the pressure of Choke tubes, Throttle Valves, Jets, Bonds etc. – At low temperature freezing can occur when the carburetor is tilted or during aerobatics in aircraft urging can occur (level of Jet distribution). In the absence of flame tropes, back firing may occur which may lead to ignition of fuel outside the carburetor.

Injection system

1. It is useful in tilted position of vehicles /aerobatics.
2. It has high volumetric efficiency.
3. It is generally less noise.
4. Weight and bulk of injection system is more than carburetor.
5. In this system complete atomization and vaporization is achieved.
6. Injection system is costly, as they precise control elements.
7. Initial cost of system is high.

Electronic fuel injection system

Modern gasoline injection system we engine sensors, a computers and solenoid operated fuel injection to meter and injection the right amount of fuel into the engine cylinder. These systems caused EFI (Electronic fuel injection) we electrical and electronic devices to monitor and control engine operation.



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Electronic control unit (ECU)

An electronic control unit (ECU) or the computer receiver electrical signal in the form of current or voltage from sensor. It then stored data to operates the injection ignition system and other engine related devices. As a result, less unburnt fuel leaves the engine as emissions and the vehicle gives better mileage. The fuel injector in an EFI is nothing but a fuel valve. When it is not energized, spring pressure makes the injection to remain closed and no fuel will enter the engine. When the computer sends the signal through injector coil, magnetic field attracts injector armature. Fuel then spurs into the intake manifold.

Direct injection system

In this type of layout fuel injection are positioned in the cylinder head to that fuel is directly discharged into each combustion chamber. – With this arrangement, it is essential that injection is timed out occur about 60° after TDC on the induction stroke. – All condensation and wetting of the induction manifold and ports in eliminated but some spray may condense on the piston crown and cylinder walls.

Indirect injection

In this arrangement the fuel is injection into air stream prior to entering the combustion chamber. Fuel spray may be delivered from single point injection (SPI) source. Which is usually just upstream from throttle (air intake side of throttle) it may be supplied from a multi-point injection (MPI) source, where the injection is positioned in each induction, manifold branch pipe just in front of inlet panel.

Common rail system

In the common rails system, a H.P pump supplies fuel under high pressure to a fuel header. high forces fuel to each of the nozzle located in the cylinder. at the proper time, a mechanically operated (by means of push and rocker arm) valve allows the fuel to enter the proper cylinder through the nozzle. The pressure in fuel header must be that for which injection system was designed i.e. it must be able to penetrate and disperse the fuel in combustion chamber. – The amount of fuel entering cylinder is regulated by varying length of stroke. A high-pressure pump is used for supplying fuel to a header, from where fuel is retired by injection (assigned one for cylinder).

CONCLUSION

In this experiment we have studied the importance of fuel injection and some details of Injection systems that are prevalent in the industry at this moment.



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Schematics

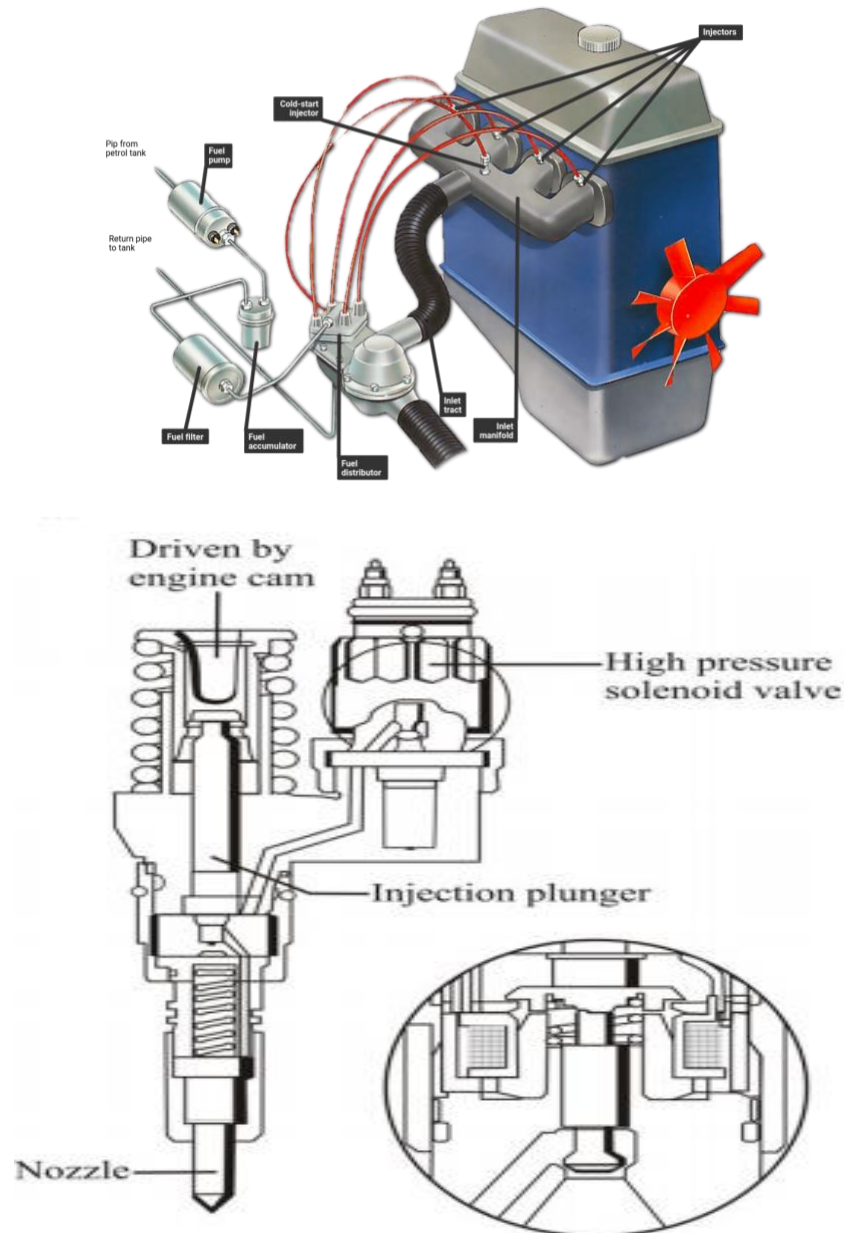


Figure: FUEL Injector

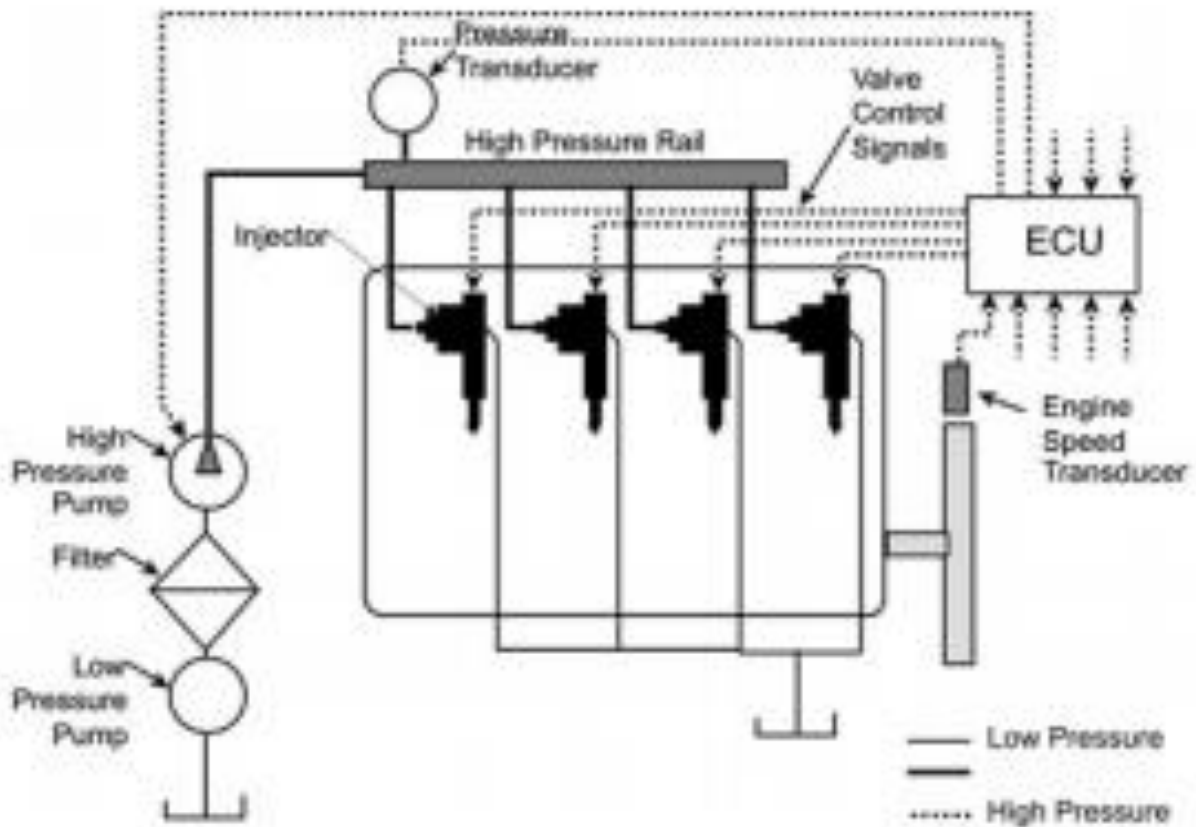


Figure: CRDI schematic



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EXPERIMENT NO. 9: TO STUDY ADVANCED VALVE TIMING SYSTEM OF A MODERN PASSENGER CAR

Object: To study advanced valve timing system of a modern passenger car.

- (i) Electronic valve timing (EVT)
- (ii) Variable valve timing
- (iii) VRTI

Theory:

Introduction: Theoretically, the opening and closing energy of the valve would occur at the crank shaft dead centers position coincident with the beginning and closing of the piston stroke.

The intake and exhaust periods of the cycle and therefore not truly intake and exhaust strokes since each includes 50 or more degrees of crank rotations in excess of a full piston stroke and the two periods actually develop to same.

Variable valve timing: In I.C. engines variable valve timing is the process of the altering the timing of valve and is often used to improve perform without VVT the valve lift timing must be the same for all engine speeds and conditions therefore common are necessary one engine equipped with a variable valve timing activation system is feed from this constitutes among performance to be improves over the engine operating range.

Cam phasing versus variable duration: The simplest form of VVT is cam-phasing when by the phase angle of the cam shaft rotated forward or backward relative to the crank shaft. Achieving variable duration on a VVT system assure a more complex system such as require multiple cam-profile or oscillating cam.

In variable valve timing engines has lower volumetric efficiency of higher sped because short time to enter the A/F mixture in cylinder as speed increasing.

Variable Shaft Timing: Variable cam shaft timing is an automobile variable valves timing technology developed by FORD.

It allows for more optimum engine performance reduced emission are increased efficiency composed to engine with fires cam shaft it weir electronically controlled hydraulic calves that direct high pressure engine oil into the camshaft power casing.

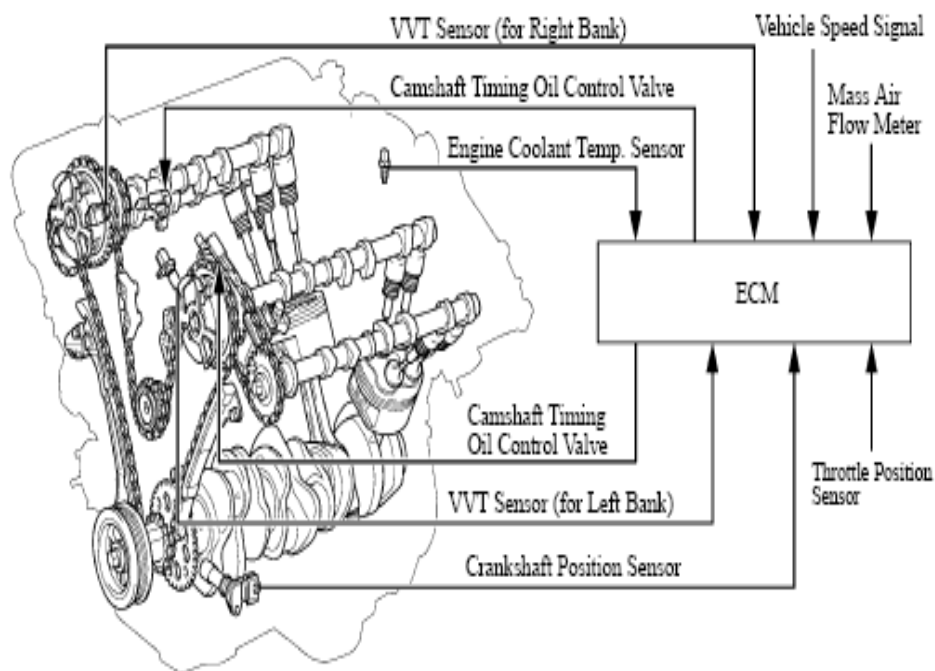
The phasor cavity changes the valves timing by rotating the camshaft slightly from its initial orientation which results in the crankshaft.

Solenoid valve: A solenoid valve is an electronically operated valve. He valve is controlled by an electric current through a solenoid in the case of a two-port valve the flow is switched on or off in the cars or a three-port valve the outflow C is switched between the two outlet ports multiple solenoid valve can be placed together on a manifold solenoid valves are most frequently used control elements in fluid.



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Variable Valve Timing with Intelligence: Variable valve timing with intelligence is an automobiles variable timing Technology developed by Taylor. VVT system is a two-stage hydraulic controlled comparing system. VVT introduced on the IZ GTE/ZIZ-GTE engine in 1996 valve the timing of the intake valve by adjusting the relationship between the camshaft drive and intake shaft. engine oil pressure applied to its conductors to adjust came position adjusting in overlap time between the exhaust valve and intake valve opening result in improve the engine efficiency.





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Experiment No. 5: To study various types of carburetor

Object: To study various types of carburetor:

1. S.U. Carburetor
2. Zenith Carburetor
3. Solex Carburetor.

Theory:

Introduction: In S.I. Engine a combination fuel air mixture is prepared outside the engine cylinder. This process of preparing the mixture is called carburetion. In this experiment we will be looking at the function of a carburetor and different types of carburetors which have made a mark in the field of Automobile engineering. The Carburetor is a device which atomizes the fuel and mixes it with air, is the most important part of the induction system. There are mainly four important factors which significantly affect the process of carburetion:

1. The time available for preparation of the mixture.
2. The temperature of the incoming air of the intake manifold.
3. The quality of fuel supplied.
4. The design of induction system to carburetor chamber.

2 Types of Carburetors

There are basically two types of carburetors, open choke and Constant vacuum type. In the former the main orifice known as choke tube or venturi is of fixed dimensions and metering is affected by varying the pressure drop across it. In the Case of Constant vacuum type, the area of the air passage is varied automatically while the pressure drop is kept approximately constant. Examples of open choke type are- Zenith, Solex Carburetor is of constant Vacuum type. Carburetor may be up draught, horizontal or down draught.

3 S.U. Carburetors

These Carburetors differ completely from open choke type being "Constant Vacuum" or "depression type" with automatically variable choke.

3.1 Construction

The Carburetor has a conventional system of that chamber which feed fuel ratio into a vertical channel in which is situated the Jet sleeve. The sleeve bears a number of holes in its sides, so that the fuel will enter into the sleeve and thus stand at the same level as in the Float chamber. Into the jet sleeve:



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1. Fits the tapered metering needle.
2. which is secured by a gid screw into the choke piston.
3. This port of the passage, one end of which is connected to atmosphere through an clearance and other to the engine through a conventional throttle valve.
4. Upper portion of choke piston which is sliding.
5. Fit fin in the suction chamber.
6. The whole assembly being located a hardened hollow steel guide rod.
7. which has its bearing in the long bar in the Centre of suction chamber.

Upper side of suction disc is connected to the air passage through a suction hole and lower side to the atmospheric through an atmospheric hole.

3.2 Working

By means of the suction hole, engine depression is transmitted to the upper face of the suction piston while atmosphere air pressure is admitted the lower chamber through the atmosphere hole. Thus, the partition of choke piston at any initial depends upon the balance of its air weight and as light spring against the vacuum force. As the weight of piston is constant vacuum choke remains constant. In unique feature of S.U. i.e., is that it has any one jet. There is no separate jet or acceleration pump. For cold starting, a rich mixture is required. This is provided by an arrangement to lower the tube away from the needle by means of jet lower, operated from data board in car. There is also an arrangement providing a slightly rich mixture in air. For this purpose, an oil dashpot (12) is provided in the upper part of hollow piston guide rod (13). A small piston is suspended by a rod. From the oil cup net. This arrangement also preserves the flatter of choke piston.

4 Zenith Carburetor

It has two concentric passages from float chamber to choke tube. The inner passage is main jet and the outer passage is the compounding jet. Main jet directly communicates with float riser. The compounding jet has a narrow restriction and the passage, the supply thrash which leads to a well subjected to atmosphere pressure. The compounding jet takes its supply from this were. Both the jets together supply almost a constant air fuel ratio during normal running of the engine. The petrol flow through the compensating jet depends upon h, difference between well and float chamber with increasing of speed. When flow becomes steady, with further increase of engine speed, flow to petrol through compensating jet remaining the same whereas air ratio increases thereby learning the main. The main purpose of Zenith carburetor is:

1. Main jet
2. Compensating jet
3. Float chamber
4. Narrow retraction
5. Well (For compensating jet supply)
6. Throttle valve



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7. Idle passage (Throttle nearby)
8. Air bleeding
9. Choke valve

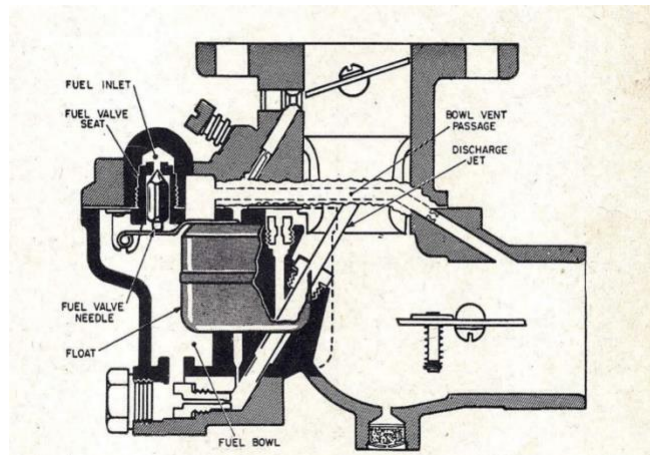


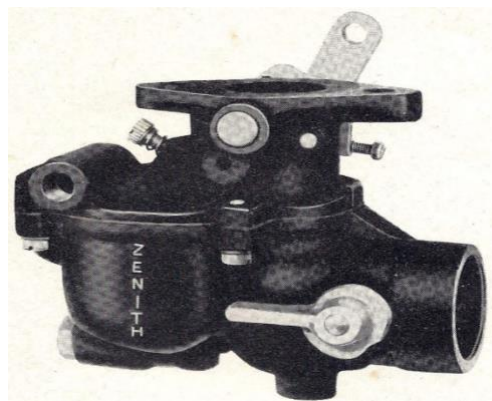
Figure 1 Section of Zenith Carburetor

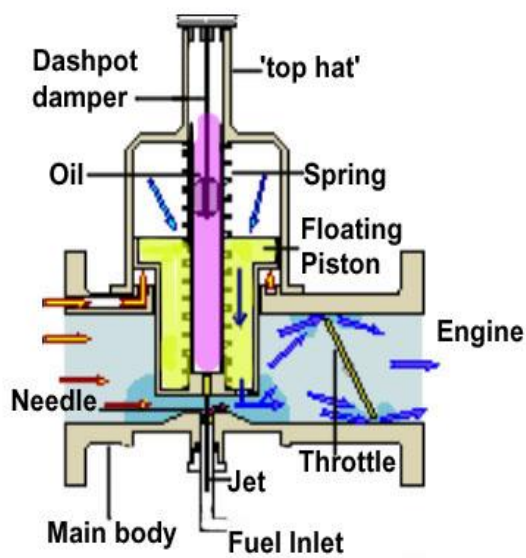
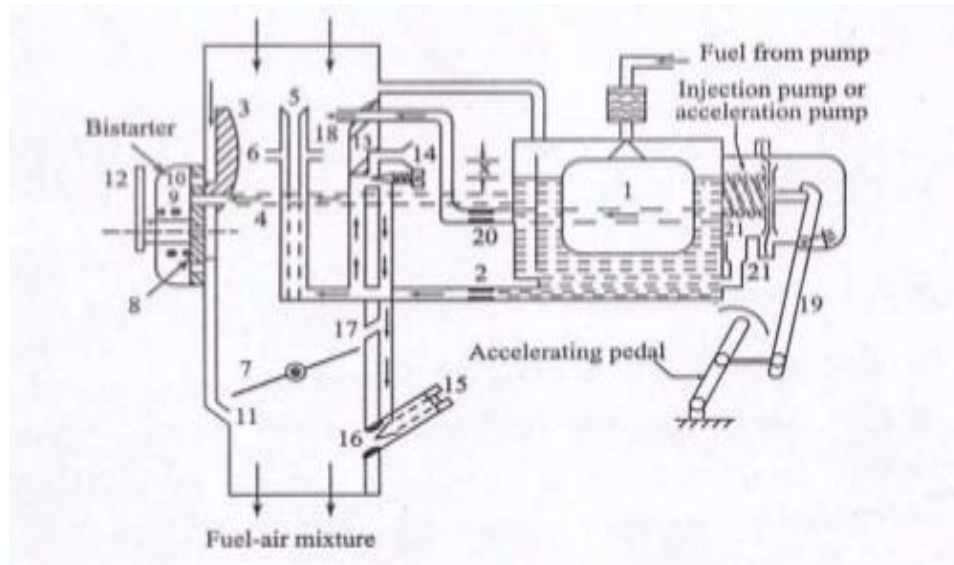
5 Solex Carburetor

Solex Carburetor – The solex carburetor is famous for ease of starting, Good performance and reliability. It is made in various models and is used in standard cons and Willy's jeep. The unique feature of this carburetor is the Bi-starts for cold starting. The various circuits of the solex carburetor are described.

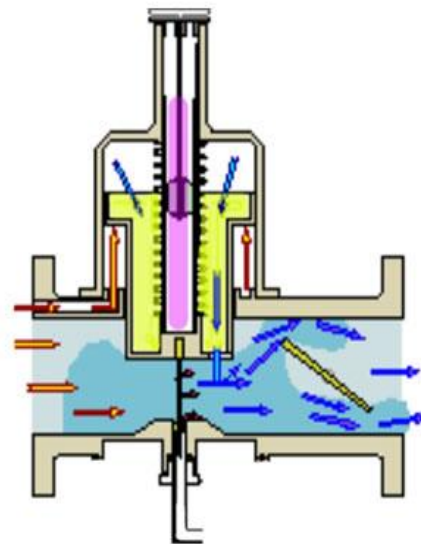
- Normal Running: The solex carburetor has a conventional float in a float chamber. For running, the fuel is provided by main jet and the air by the choke tube or venture. The fuel from the main jet passes into the well of the air bleed emulsion system.
- Cold starting and warming: The unique feature of solex carburetor is the provision of a Bi-starter or a progressive starter. The starter valve is in the form of a float disc with holes of different sizes.

Idling and slow running: From the lower part of the well of the emulsion system a hole leads off to the pilot jet. At idling the throttle is almost closed and engine suction is applied at the pilot petrol jet.



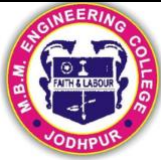


SU at idle



SU at open throttle

Figure 2 S.U. Carburetor



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Experiment No. 3: DC generator and battery charging methods and apparatus.

Object – To Study DC Generator and alternator storage battery charging drive

Theory – Generator is a device to convert mechanical energy the generator replaces in the battery

- ⇒ The current is used in starting the engine and also supplies current for operation system light ratio etc.
- ⇒ The automobile generator is usually a also voltage DC generator reviver DC

Principle of Generator – When a conductor is moved in a magnetic field. A current is produced in it. The conductor must have across the magnetic field so that it acts through live of force. Voltage developed; $e = bLV \times 10^{-8}$ volt

$$e = N Z / 60, \quad /A \text{ volt}$$

where $N = \text{rpm}$

$Z = \text{No of conductor}$

$= \text{pole strength (flor density)}$

$A = \text{no. of parallel parts}$

$P = \text{No. of pole}$

Construction of DC generator – It consist of pole pieces fitted in a frame, current are commutator and field winding.

- ⇒ Pole pieces are laminated iron cores for the field winding that famish the magnetic field and generator.
- ⇒ Amative is made up conductor of injected are around on a laminated iron cave rotating anatine without circuit

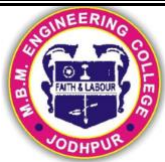
Cut out relay – The cutout relay closed the circuit between the generator and the battery when the generator is frothing current.

DC generator is converted to the battery through cut out relay and an ammeter

- ⇒ It is singly a magnetic switch. the aromatic mounded a hinge above the core. The amative is connected to the battery through town an ammeter by contact point.
- ⇒ When the generator is operating the core winding creoles magnetic field which alternate the amative contact pome are adored and generator is cone ailed to the battery and faces from the series winding. the this open the circuit between the generator and the battery.

Battery – The battery is the main part of electrical system in an automobile. the battery supplies current for operation of the stating mole and ignition system when the engine is being cranked for storing

It is of following type –



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- (1) Lead acid battery
- (2) Alkaline battery
- (i) Ni – Fe type
- (ii) Ag – Zn type
- (iii) Ni – Cd type
- (3) Zine –air battery
- (4)

Lead acid battery – It consist of following components: -

- (1) Container
- (2) Plates
- (3) Separator
- (4) Electorate
- (5) Temirass
- (6) Cell covers
- (7) Grid

Container – It is a single piece container and is made of hard rubber plastic or bituminous compotrenres. It is divided into comportment by partitive for differed curs.

Plates – several similar plates are property spaced and welded or lead bumed strop to from plate group. The grids are on ally of lead and antimony which makes them resistant to electrode corrosion.

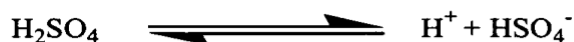
Grids: These grids serve to conduct the current to and from the active material of the positive and negative plates. An alloy consisting essentially of lead and antimony is used for the grids.

Separators: They are made up of specially treated wood, hard rubber, rain impregnated fiber alloy or in combination with rubber or of glass fiber.

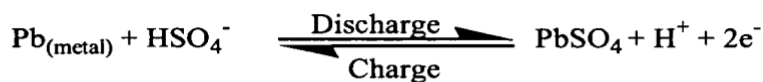
Electrolyte: The electrolyte of a fully charged battery contains about 31% sulphuric acid by weight or about 21% by volume in distilled water.

Chemical Reaction:

Equation 1
Electrolyte



Equation 2
Negative Electrode



Equation 3
Positive Electrode



Equation 4
Total Reaction



When battery is charged:

The lead sulphate ore plate is again converted to PbO_2 ; Pbso_4 on other plate is reduced to spongy lead.

The chemical energy into electrical energy during discharging.

Chemical Responsibility: The most valuable characteristic of the lead acid battery is its chemical reversibility. It means that unlike a dry cell battery which must be thrown away when it becomes discharged, the strong battery may have an electrical current paired through it in the direction of charge.

Discussion:

Altimeter are used instead of DC generator as:

- (1) Size is reduced.
- (2) No current is reduced.
- (3) Lighter in weight.

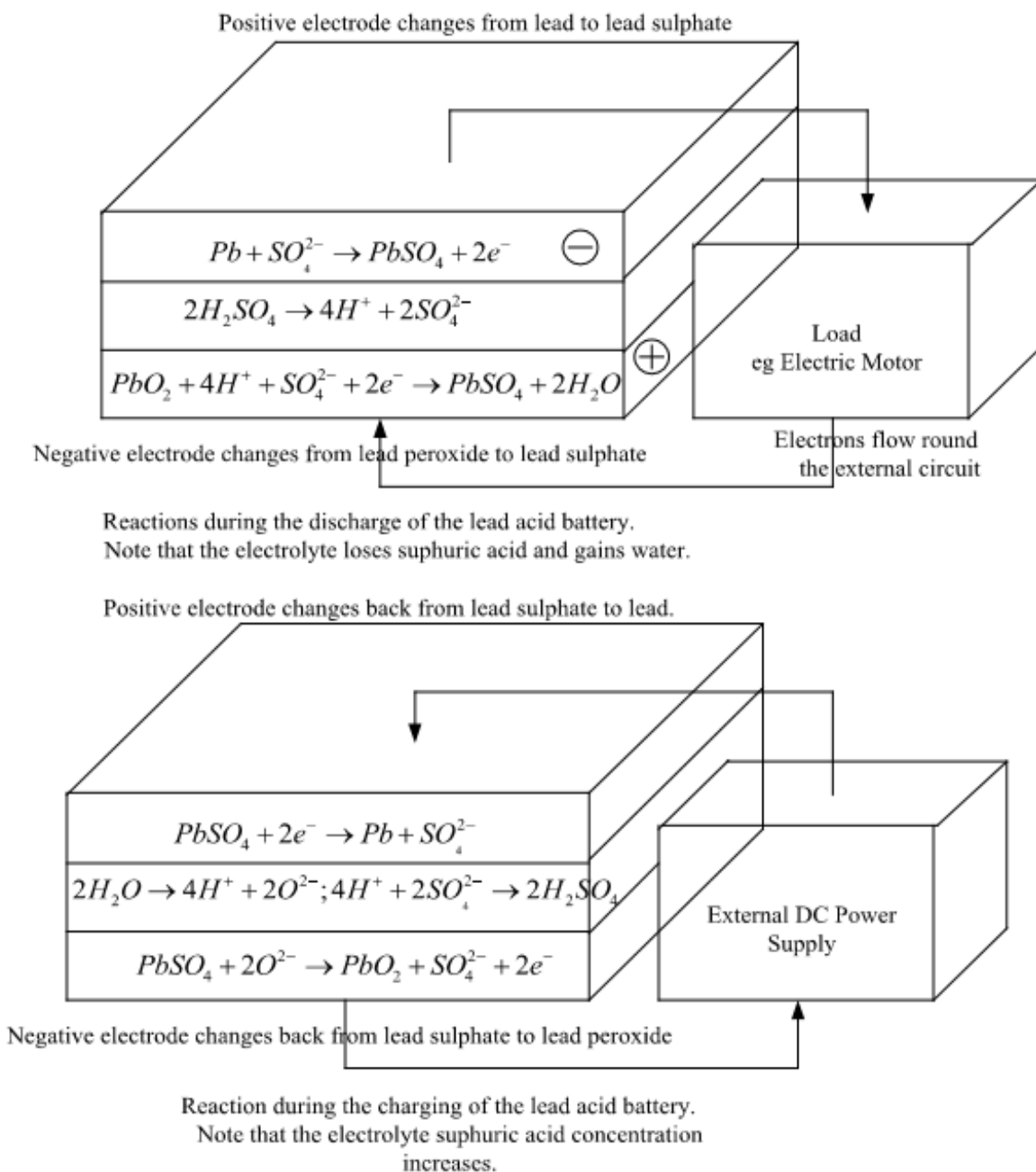
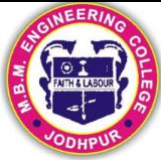


Figure 3 Charging and discharging of Pb- acid battery



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Experiment No. 2: To study Engine starting motor (D.C. Motor) and driving arrangement (Benedix Drive).

Theory — Internal combustion engines are feedback systems, which, once started, rely on the inertia from each cycle to initiate the next cycle. In a four-stroke engine, the third stroke releases energy from the fuel, powering the fourth (exhaust) stroke and also the first two (intake, compression) strokes of the next cycle, as well as powering the engine's external load. To start the first cycle at the beginning of any particular session, the first two strokes must be powered in some other way than from the engine itself. The starter motor is used for this purpose and it is not required once the engine starts running and its feedback loop becomes self-sustaining. In this Experiment we will be looking at the basics of Benedix drive, a type of engagement mechanism used in starter motors of IC engine. and D.C. Starter motor.

I.C. engines are not self starting and need to be rotated at a certain minimum speed in order for the engine to commence running by the fuel supply. This is the function of the starting motor. It cranks the engine for starting, Cranking the engine means to rotate the crankshaft by applying torque on it so that the piston may get reciprocating motion. the starting motor is mounted on the engine flywheel housing. The slower it turns, the more torque it develops. The function of motor is to develop a torque or turning effect when the current is passed through the conductor according to Fleming's right hand rule. Benedix Drive is a device that allows the pinion gear of the starter motor to engage or disengage the ring gear (which is attached to the flywheel or flex-plate of the engine) automatically when the starter is powered or when the engine fires, respectively. It is named after its inventor, Vincent Hugo Bendix.

2 Motor working principle

: The principle of an electric motor is based on the current carrying conductor which produces magnetic field around it. A current carrying conductor is placed perpendicular to the magnetic field so that it experiences a force.

3 Drive Arrangement

: The starting motor is linked to engine through a set of gears. A pinion gear is attached to the starter armature which drives a ring gear, attached to the flywheel.

The arrangement is made that the two gears engage, to crank the engine until it starts and then disengage automatically when the engine is running. The Drive arrangement may be of two general type:

1. Inertia drive (benedix drive).
2. Over-running Clutch.

4 Benedix Drive

It is fastened on the armature shaft of the starting motor. The drive head is keyed to the end of the armature shaft. The pinion gear having internal thread is mounted on the thread sleeve. Just like a nut on the bolt. A spring is attached to the drive head and also to the sleeve. When the starting motor is at rest the pinion gear is not engaged with the flywheel. When the starting motor is switched on, the armature begins to rotate. This causes the sleeve to rotate also. Until it engages with the teeth of the flywheel. The slight turning of the pinion gear helps to engage it properly with the flywheel. The spring between the armature shaft and the threaded sleeve takes the shock of the starter. After the engine has started, the pinion gear is turned by the engine much faster than when rotated by the starting motor. This causes the pinion gear to turn back on the threaded sleeve making it disengaged with the flywheel.

5 New technology

Some modern gasoline engines with twelve or more cylinders always have at least one or more pistons at the beginning of its power stroke and are able to start by injecting fuel into that cylinder and igniting it. If the engine is stopped at correct position, the procedure can be applied to engines with fewer cylinders. It is one way of starting an engine of a car with stop-start system.

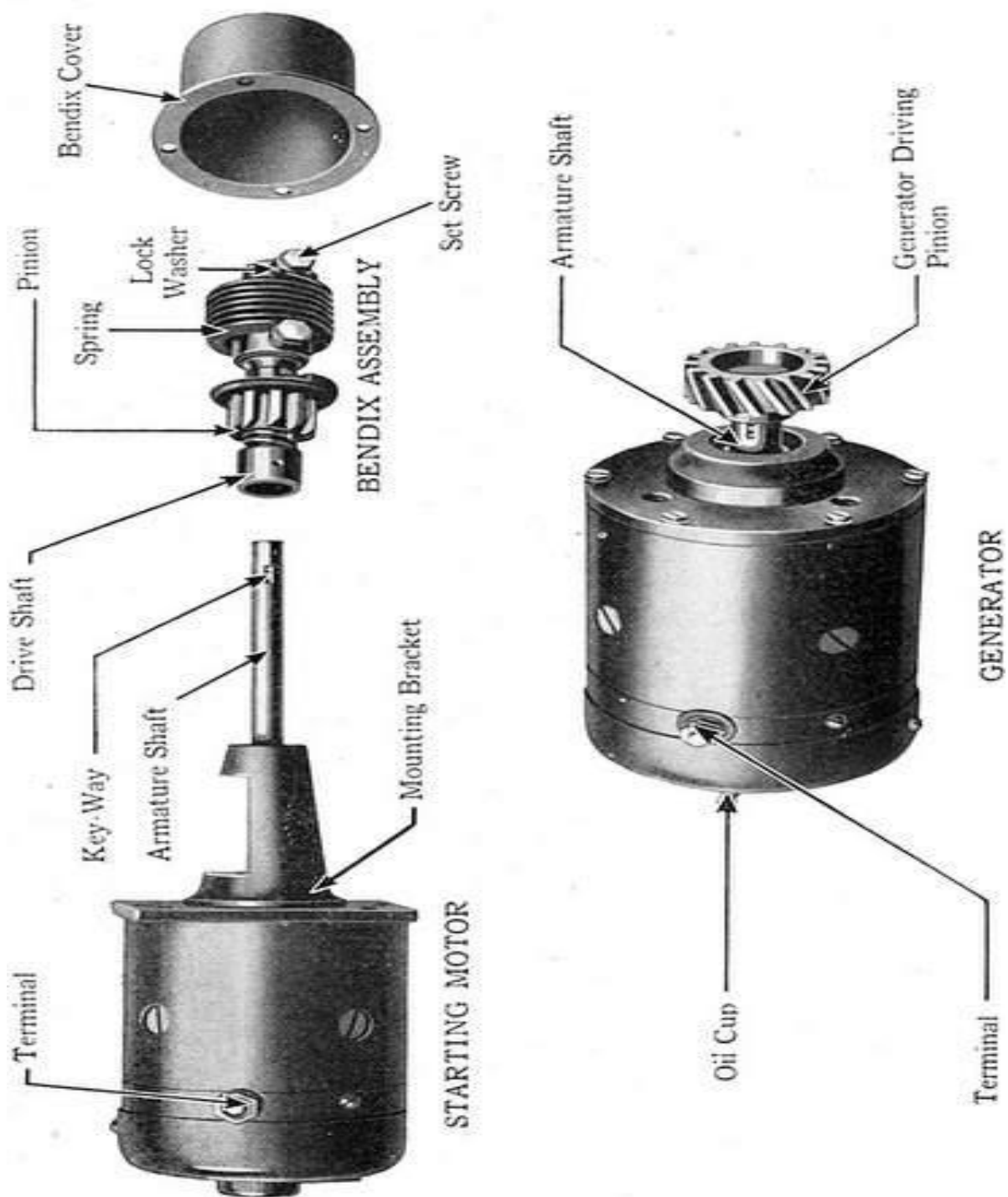


Figure 4 Bendix disassembled



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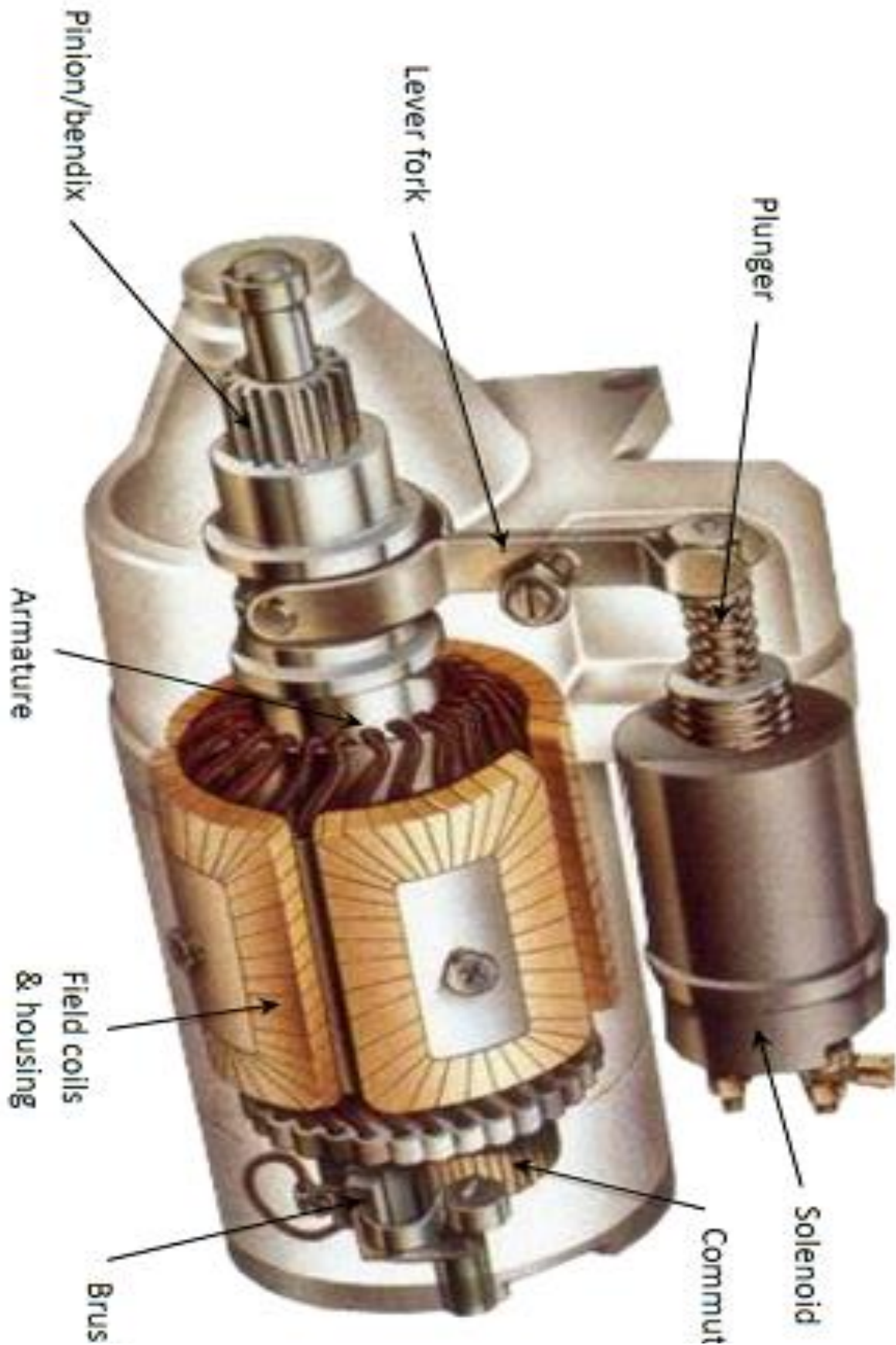
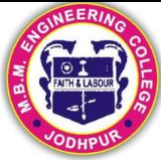


Figure 5 Bendix Drive arrangement X-Sc



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Experiment No. 10: Morse Test on SI engine

Object: To perform morse test on petrol engine and draw the performance curves.

Theory

Introduction: The difference between indicated and brake power of an engine is known as Friction power. The internal layer is an engine are efficiently of two kinds viz, pumping layer and friction layers. Following methods are used to find friction power to estimate performance of the engine.

Morse test: the morse test consists of obtaining indicates power of the engine without any elaborate. The test consists of making each cylinder of the engine and nothing the reduction in break power developed with a gasoline engine each cylinder is retendered inoperative by showing the spark plug or the cylinder with a different engine by cutting off the supply of fuel to each cylinder.

When one cylinder is at brake power is

$$B_1 = B - I_1 \quad (1)$$

When second cylinder is cut brake power is

$$B_2 = B - I_2 \quad (2)$$

When 3rd cylinder is cut off Brake power is

$$B_3 = B - I_3 \quad (3)$$

When 4th cylinder is cut off Brake power is

$$B_4 = B - I_4 \quad (4)$$

When I_1, I_2, I_3 , and I_4 are indicated power in cylinder I, II, III, IV

$$B_1 + B_2 + B_3 + B_4 = 4B - (I_1 + I_2 + I_3 + I_4)$$

$$I_1 + I_2 + I_3 + I_4 = 4B - (B_1 + B_2 + B_3 + B_4)$$

Willian's Line Method: This method is also known as fuel rate extrapolation method. A graph commencing fuel consumption a brake power at constant speed is drawn and it is extrapolated on the negative axis i.e. taken as the friction power or the engine at that speed indicated power is that sum of the brake power and friction power.

Motoring Test: In motoring test engine is steadily operated at rated speed by its own power and allowed to remain under given speed and load condition for sufficient time to that temperature of each engine. Component reaches a steady state. An electronic dynamometer or weld to absorb power during this method period. The ignition is the cut off and by suitable switching devices. Dynamometer is converted to run as a motor to as to crack the engine at some speed at which it was operating the power supply from above dynamometer is measured which is a measure of primitive power of engine.

Indication Diagram: The device which measure variation of pressure in cylinder are a pore air cycle is caused indicator are plot or such information obtained is caused an indicator diagram developed and shaded are represents pumping power. Therefore, indicated power is

$$IP = bP - PP$$

In practice PP is generally ignored since it is very small.

Thus the area of indicative diagram is accurately measured which represents indicated power of the engine.



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TABULAR COLUMNS:

Sl. No.	Speed, rpm	Spring balance Wkg	Manometer Reading			Time for 10 cc of fuel collected, t sec
			h_1 cm	h_2 cm	$h_w = (h_1 - h_2)$	

MORSE TEST:

Sl.No	Cylinders ON	Cylinders OFF	Load Kgs	Speed (N)rpm	BP
1					
2					
3					
4					
5					

CALCULATIONS:

1. Area of Orifice $A_o = \frac{\pi}{4} d_o^2$ cm² (d_o is orifice diameter = 25mm=0.025m)

2. Head of Air $H_a = \frac{(h_1 - h_2) \times \rho_w}{\rho_a}$ (in mts; $\rho_w = 1000 \text{ kg/cm}^3$)

$\rho_a = 1.2 \text{ kg/cm}^3$, h_1 and h_2 in mts

3. Mass flow rate of Air M_a in kg/hr

$$M_a = A_o \times C_d \times 3600 \times \rho_a \times \sqrt{2 \times g \times H_a} \text{ kg/hr}$$

4. Total fuel consumption TFC : in kg/hr

$$TFC = \frac{10 \times 3600 \times \rho_f}{t_1 \times 1000}$$

5. Brake Power BP in Kw

a. With hydraulic brake dynamometer (reaction type)



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b. $BP = [2 \times \pi \times 9.81 \times N \times W \times R] / 60,000 \text{ kW}$

Where $R =$ Load arm length = 0.3mts
 $W =$ load shown on spring balance, kg
 $N =$ speed in rpm

6. Specific fuel consumption: SFC in Kg/Kw-hr

1. $SFC = TFC / BP$

7. Air Fuel ratio : A/F

$$A/F = Ma / TFC$$

8. Brake Thermal efficiency

$$\eta_{bth} = [BP / TFC \times CV] \times 100\%$$

With MORSE TEST:

- Determine B, B1, B2, B3 and B4 – Brake powers as above
- Indicated Power in kW, $IP = [4 \times B] - [B1 + B2 + B3 + B4] \text{ kW}$
- Mechanical Efficiency $\eta_m = BP / IP$

9. Indicated Thermal efficiency

$$\eta_{ith} = [IP / TFC \times CV] \times 100\%$$

GRAPHS:

Plot curves of BP vs. TFC, SFC, A/F, and Mechanical efficiency.

MOTORING TEST:

CALCULATIONS:

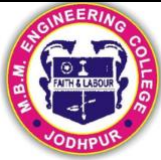
1. FRICTION POWER, FP

$$FP = (V \times I) / 1000 \text{ KW}$$

Where,

$V =$ voltmeter reading on motoring side

$I =$ ammeter reading on motoring side



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Experiment No. 7: Lubrication Systems

Object:

- To study various lubrication systems:
 1. Splash
 2. pressure feed
 3. mechanical
- To study lubrication of Piston rings.

Theory:

Lubrication is essentially required in motor vehicle maintenance. To supply lubricating oil between the moving passage is simply termed as lubrication.

Lubrication of all moving parts (except nylon, rubber, lubricated components) is essential to reduce friction, wear and prevent seizure.

Objects Of Lubrication:

1. to reduce friction between moving parts
2. To reduce wear of all moving parts
3. To act as cooling medium for projecting heat next line to keep the engine pass clean, especially piston rings, ring grooves, oil ways, filter and etc.
4. To absorb shock between bearing and other engine parts.
5. Reducing engine noise is an extending engine life next line to form a good seal between piston rings and cylinder walls
6. To prevent deterioration of rubbing objects
7. To prevent absorption of carbon, suit as lacquer.
8. To absorb and carry away harmful substances resulting from incomplete combustion.
9. to prevent metallic components from corrode corrosion attack by acid formed during the combustion process.
10. To resist oxidation which causes sludge and lacquer.

The engine lubrication system may be either pressure type or splash type although various combinations are used a single system lubricates the entire engine of all present day motor vehicles. The pump is located at the bottom of the crankcase. Next line in the splash system, the connecting rod end has a dripper which splashes the oil on the vehicle pass as it traverses through the oil while moving in the connecting Rd.

in **splash system**, the connecting rod has a dripper which splashes the oil on the various parts.

Splash System: In this system of lubrication the lubricating oil is stored in an oil some. A scoop or dripper is made in the lowest part of the connecting rod will stop when the engine runs the tripper dips in the oil once in every revolution of the crankshaft and causes the oil to splash on the cylinder walls.

Splash system mostly works in connection with pressure system in an engine some parts being lubricated by **splash system** and other by pressure system.



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Pressure System: In this system of lubrication the engine parts are lubricated under pressure feed. The lubrication oil is stored in a separate tank or the some, from where an oil pump takes the oil through a strainer and deliver it to the filter to the main oil Gallery at a pressure of 4 to 8 N/ cm square.

from the crank pin it goes to the piston pin through a hole in the connecting rod, web where the lubrication of piston ring takes place.

The wall tappets are duplicated by connecting the mean oil Gallery to the tappet guide surface is through the divided holes will stop

An oil pressure gauge and the instrument panel indicate the oil pressure in the system will stop oil filters and strainers in the system cooler of oil from dust, metal particles and other harmful particles.

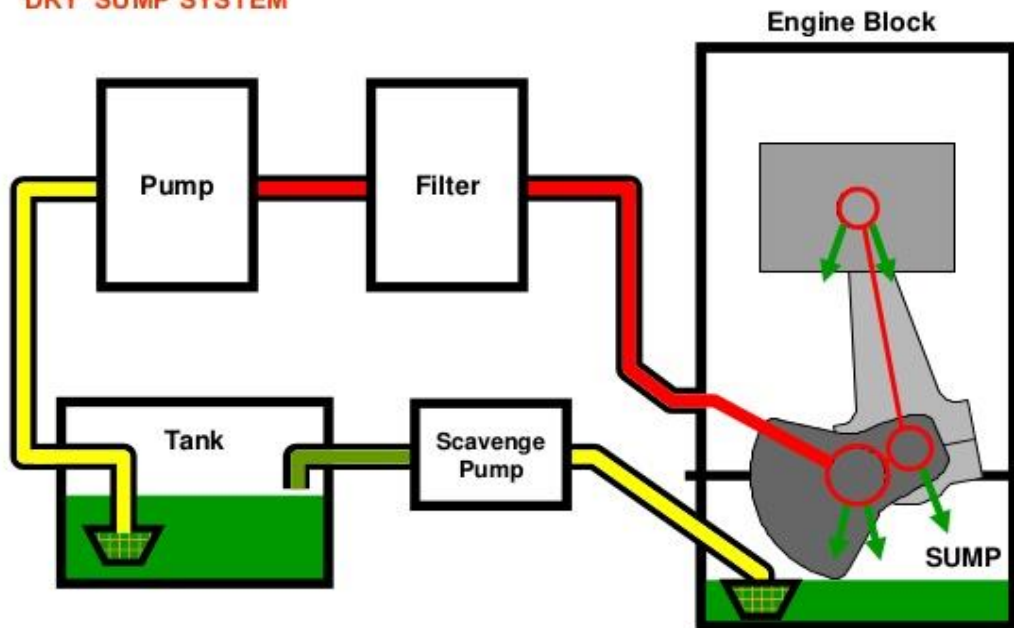
Automatic lubrication: It is a device fitted to a steam engine to supply lubricating oil to the cylinders and sometimes the bearing an Excel box mounting as well these are various types of automatic lubrication which include various designs of displacement, hydrostatic, mechanical lubrication

Ring oil system: This system of lubrication is more useful for medium speed engines because at high speeds the oil will be thrown off due to centrifugal force and at low speeds the amount of oil carried will be inadequate.

Drop feed lubrication: Drop feed lubrication or drop feed oil lubrication contains a oil Cup. It contains of an oil retainer or cub made of glass could stop at the top of the Cup there is an opening called filling hole through which the lubrication oil is poured into the Cup of feed glasses provided to see the drops of the oil.

Schematics:

'DRY' SUMP SYSTEM



High Performance/Rally vehicles and Aircraft

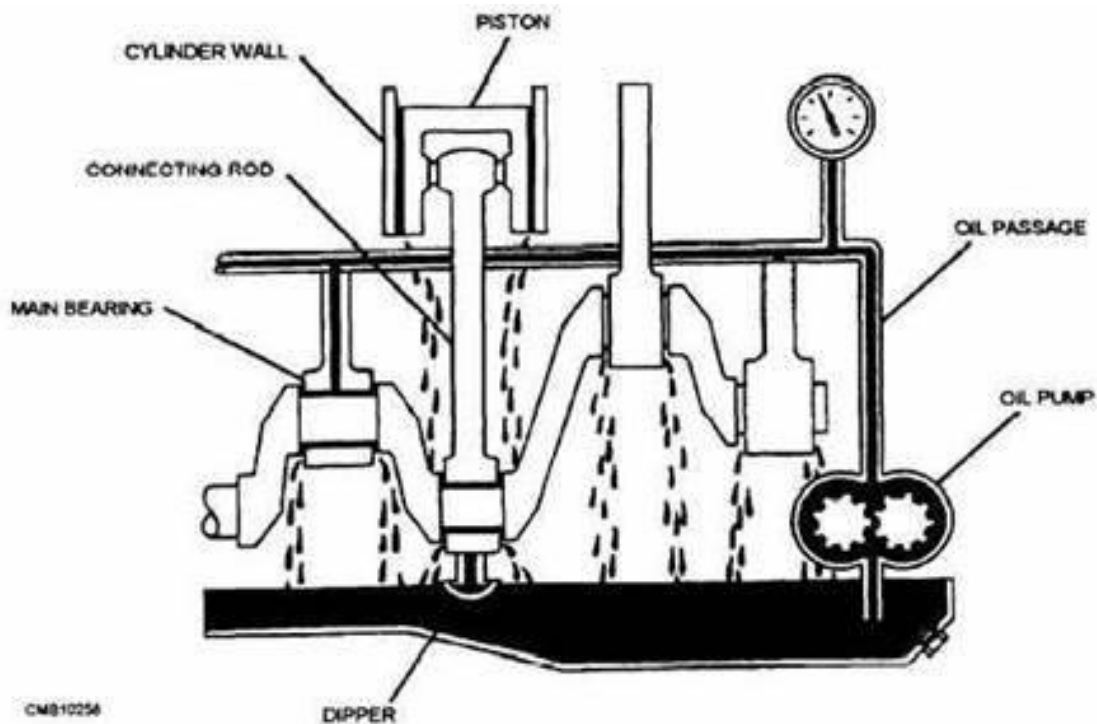
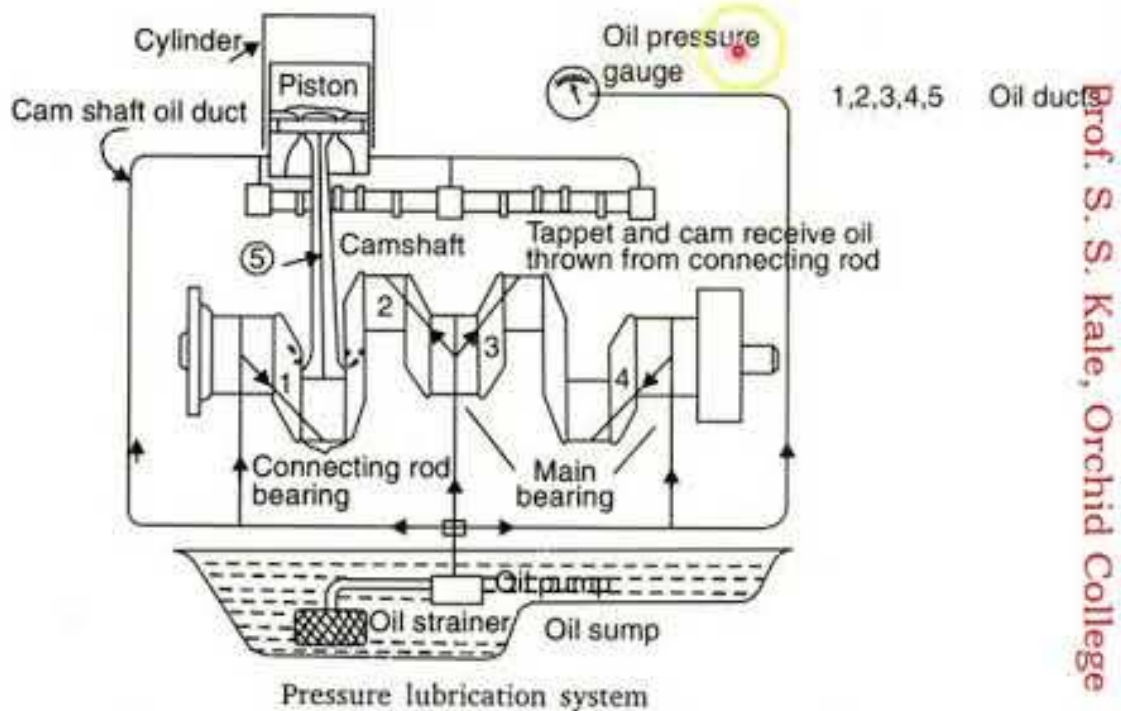


Fig2.) Splash Lubrication

Pressure Lubrication System



Prof. S. S. Kale, Orchid College

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 Car engine lubrication system

Fig3.) Pressurized Lubrication System



MBM Engineering College Jodhpur Department of Mechanical Engineering

Experiment No. 8: Ignition Systems

Object:

- To study advanced ignition systems for SI engine.
 - Battery ignition system
 - Magneto ignition system
 - Transistorized arrested contact ignition system
 - Capacitive discharge ignition system
 - Electrolytic ignition system
 - Piezo electric ignition system

Theory:

Ignition system: Ignition system generates a spark or heat on an electrode too high temperature to ignite the fuel air mixture in spark ignition internal combustion engines oil fired and gas fired boilers rocket engines etc under compression in the engine cylinder.

Battery ignition system: A conventional coil ignition system or battery ignition system involves a coil in a primary or low turn terminal about \$200. Off FireWire, all wrapped around an iron core however all these systems have the coil in the same way as the contact breaker but are less prone to wear and misalignment without the mechanical limitations of a contact breakage coil ignition system can produce up to 500 Sparks per second.

Magneto ignition system: In a Magneto ignition system there is no need for a battery since a current is induced in the Lt winding by charging it in the magnetic field. Magneto ignition is best suited to engine that are independent of a battery will stop A Magneto supplied ignition system voltage for spark discharge independent of a battery and commonly used in small 4 stroke and two stroke engines. Operation a time varying magnetic flux of high voltage is set up in the ignition armature as the rotating permanent magnets on the pole in generate a current in the primary winding. This primary current generates on additional flux 51 giving a result in flux R. $R = O + I$. The current can be interrupted with contact Breakers.

Difference between battery and Magneto ignition system

Battery ignition system:

1. bulky and scattered
2. Maintenance is easier.
3. Cost of manufacturing is cheaper

Magneto ignition system:

1. Light and compact
2. Maintenance is bit difficult
3. Cost of manufacturing is easier



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Drawbacks of conventional battery / Magneto ignition system:

Excessive of contact breaking points, handle of inductive current interruption effect the line of contact breaking points, life of point reduces if condenser is weak.

The arm of contact breaking points bounces of high speed at low speeds that well. Is more more current is drawn from the battery slow operating of contact breaking points hence low secondary rotating.

Short life of breaking point.

Transistor arrested contact ignition system:

TACIS provided high voltage and and the electronic triggering to maintain required timing as four neglecting conventional ignition system. Emitter E is connected to ignition system switch and collector C to primary through battery coil.

The current flows through contact breaker is reduced by a large amount due to fact that contact breaker has to switch only based current of transistor which is 1/B part of that collected current this improves the life of contact breaker.

Advantages:

- Ben reduced ignition systems maintenance.
- Reduced wear of components
- Increased reliability
- Increased efficiency.

Capacitive discharge Ignition System:

The CDI works by first up the battery voltage to about 400 volts DC ring on oscillation. This voltage has a very fast rise in time compared with a more conventional system.

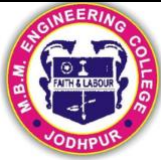
Typically the rise of time for CDI is 3 to 10KV per millisecond.

this very fast rise and high voltage with ensuring that even a oil fouled plugs will be fired however when used in complement with direct ignition that spark duration is expectable.

Electronic ignition system:

Electronic ignition system is now fitted to almost all spark ignition vehicles this will cause the conventional spark ignition mechanical system has some maintenance disadvantage.

1. mechanical problems in the contact breaking not the least of which is the limited lifetime but stop
2. Current flow in the primary circuit is limited to about four amperes could stop
3. Damage will occur to Contacts at least in the lifetime will be seriously reduced. Next Lane legislation requires straight emission limits, which means the ignition must stay in time for a long period of time.
4. Weaker maintenance requires more energy from the spark to ensure successful ignition even when high engine speeds are achieved.



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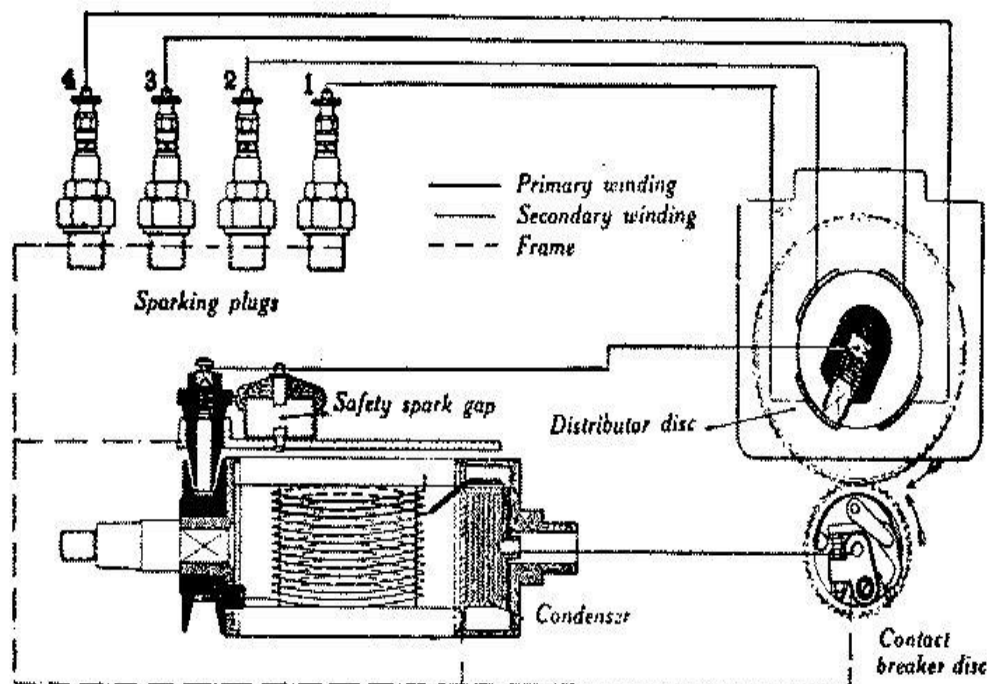
Piezo Electric Ignition System:

Diesel ignition system is a type of ignition that is used in portable camping stoves, gas grids, lonely ignitions good stop

This consists of a small spring-loaded hammer which when hence the piezo crystal.

1. This is the most recent system.
2. Can crystal of borium titrate and lead zirconate R subjected to compression it generates electric voltage used in gramophones an unicrophones.
3. when pressure is applied off 500 bar voltage generated is about 19KV.
4. The system used in USA develops 20K hold at a pressure of 400 Newton per cm square.
5. so each box at low speed is starting and high speed of engine.
6. It can be used with fused plunger.

SCHEMATICS:



(CONTACT) BATTERY IGNITION SYSTEM